

Arunabha Majumder

Mechanical Designer | 3D CAD & Technical Drawings | New Product Development | Prototyping

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PROFESSIONAL SUMMARY

Mechanical Designer with a B.Tech in Mechanical Engineering, an M.Tech in Mechatronics and about three years of hands-on new product development in Denmark. Turns concepts and requirements into practical electromechanical designs: 3D CAD models and technical drawings in SolidWorks, FEA-driven iterations for stiffness, strength and weight, and prototypes built, assembled, tested and verified hands-on. Brings a structured approach with attention to detail, keeps cost and manufacturability in mind, and collaborates closely with engineers across disciplines. Certified SolidWorks Associate (CSWA), fluent in English (C1), based in Aalborg.

CORE SKILLS

- **3D CAD & Drawings:** SolidWorks (CSWA certified), Fusion 360, 3D CAD modelling, assemblies, creating and interpreting technical drawings, Finite Element Analysis (FEA)
- **Mechanical Design:** New Product Development (NPD), concept to tested prototype, Design for Assembly (DFA), design for manufacturability, cost and part-count reduction
- **Manufacturing:** Machined parts, additive manufacturing / 3D printing, prototype fabrication and assembly
- **Prototyping & Verification:** Prototype development, testing, product verification, custom test rigs, experimental validation
- **Electromechanical Systems:** Electromechanical products: motor, encoder, force-sensor and DAQ integration, actuator design
- **Ways of Working:** Structured design documentation (3D models, drawings), cross-functional collaboration, ownership, English (C1)

PROFESSIONAL EXPERIENCE

Research Engineer – Mechanical Design & Product Development (PhD Project: VIEXO) | 2023 – Present

Aalborg University, Aalborg, Denmark

- Took a wearable electromechanical product from concept and requirements to tested hardware, owning the SolidWorks 3D CAD models, assemblies and technical drawings throughout development.
- Iterated designs on FEA results for stiffness, strength and weight before fabrication, and validated analytical stiffness models against physical tests to below 10% RMS error.
- Built prototypes hands-on from machined and additively manufactured parts, assembled them, and integrated motor, encoder, force sensors and DAQ.
- Planned and ran testing and verification with custom automated test rigs, including a 12-participant study across 3 assistance modes and 2 load conditions.
- Reduced system cost, mass and part count of a 5-bar parallel robot by removing the external force/torque sensor, using a force-estimation method derived from the actuator itself.
- Cut energy consumption by more than 25% with a hybrid actuation design combining a parallel spring and a 6 Nm motor; awarded Best Paper, IFToMM ISRM 2026.
- Taught university students and supervised Bachelor's (B.Tech) students (2023 – 2025).

M.Tech Researcher – Actuator Systems (Master's Thesis) | 2020 – 2022

CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

- Fabricated pneumatic artificial muscle actuators in-house and integrated valves, pressure sensors and DAQ to characterise load–displacement behaviour and derive an empirical actuator model.

- Reduced dynamic position-tracking error to 0.3–0.78% with a neural-network-based gain-scheduled PID controller for a non-linear pneumatic actuator.

Mechanical Engineering Capstone – Upper-Limb Rehabilitation Exoskeleton | 2018 – 2019

Siddaganga Institute of Technology, Tumakuru, India

- Led CAD design, fabrication and user testing of a rehabilitation exoskeleton; won the Best Major Project Award and an INR 100,000 TATA Technologies Innovation Grant.

EDUCATION

Ph.D. Candidate, Mechatronics & Robotics | 2023 – Present

Aalborg University, Department of Materials and Production, Aalborg, Denmark

M.Tech, Mechatronics | 2020 – 2022

AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

B.Tech, Mechanical Engineering | 2015 – 2019

Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India

ENGINEERING PROJECTS

- **Go-Kart (2017):** Led end-to-end design and fabrication of a custom Go-Kart built from the ground up.
- **Efficycle (2018):** Worked in a multidisciplinary team to engineer a human-electric hybrid vehicle for the Efficycle competition.

CERTIFICATIONS & AWARDS

- Certified SolidWorks Associate (CSWA) – Mechanical Design
- Best Research Paper Award, 9th IFToMM International Symposium on Robotics and Mechatronics (2026)
- Best Major Project Award, Department of Mechanical Engineering (2019)
- TATA Technologies Innovation Award – INR 100,000 Grant (2019)

PUBLICATIONS

- Majumder, A., Wagner, J.W., Zhu, Y., Oliveira, A.S., and Bai, S. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." *Robotics and Mechatronics: ISRM 2026, MMS*, vol. 213, Springer.
- Majumder, A., Sarkar, D., Chakraborty, S., Singh, A., Roy, S.S., and Arora, A. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." *IEEE CONECCT*, 2022.

TECHNICAL TOOLS

- **CAD & Manufacturing:** SolidWorks, Fusion 360, FEA, Bambu Lab and Ultimaker Cura (3D printing)
- **Software:** MATLAB/Simulink, Python, LabVIEW (basic), Arduino
- **Testing:** Qualisys Motion Capture, Forsentec Force/Torque Sensors, Custom Rig Fabrication

LANGUAGES

English (C1, fluent) | Bengali (Native) | Hindi (Conversational)

REFERENCES

Available upon request.